PSVM + lab → P.S.V.M.

Java Operators

1. Arithmetic operators (+, -, *, /, %
+=, -=, *=, /=, %=
++, --)

2. Relational operators : (==, !=, <, >, <=, >=
return boolean values

3. Bitwise operators
↳ Bit Manipulations (0 or 1)
(&, |, ^, ~, >>, <<, >>>, &=, |=, ^=,
>>=, <<=, >>>=)

Truth table.

A	B	A & B	A | B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

shift operators (>>, <<)

1) Left shift (<<)

byte b=8;

b = b<<1

128	64	32	16	8	4	2	1	
0	0	0	0	1	0	0	0	= 8 × 2
0	0	0	0	1	0	0	0	= 16 × 2
0	0	0	1	0	0	0	0	= 32

shift every bit one position, left

leftshift ≈ × 2

If 8-bit number → 6th bit (+ve)
→ 7th bit (-ve)

n-bit no → (n-2) bit (+ve)
(n-1) bit (-ve)

If we shift 30th time → normal value
31th time → ~~negative value~~
32 time = no. % 32 ⇒ remainder

Type promotions → <<, >>
↳ int, long (✓)
↳ byte, short, (✗)

if you do in byte or short, then first it will type promote to int and then perform in int.

Note - bitwise always performed on
↳ int long ✓

Right shift

byte b = 8;
b = b>>1
(byte)

0	0	0	0	1	0	0	0	
128	64	32	16	8	4	2	1	
0	0	0	0	1	0	0	0	= 8
128	64	32	16	8	4	2	1	
0	0	0	0	0	1	0	0	= 4

MSB

>>> (shift Right with zero)

byte b = -128

1	0	0	0	0	0	0	0	= -128
1	1	0	0	0	0	0	0	
0	1	0	0	0	0	0	0	= +64

b>>1;
b>>>1 → put zero

makes negative no to positive by placing zero

4. Logical operator
↳ && (AND) (if one false then ⇒ false) (both exp true → true)
↳ || (OR) (if one true & other false → ~~both~~ true)
↳ ! (NOT) (optional)

works on expressions not on bits.

Short circuit

we use (A && B) condition when if we don't need to check another condition if 1st condn is false.

same for (A || B) if 1st condn becomes true, then don't check other condn

In Bits

A	B	A & B	A | B
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

logical

A	B	A && B	A || B
F	F	F	F
F	T	F	T
T	F	F	T
T	T	T	T

In Java we can use Bit operator in expressions also, not only for bits.

Inverted condition

eg : d = (a < b) & (a < c)

works same as && → but doesn't do short circuiting

If we don't want short circuit → Use & or | (unary bitwise operator)

5. Assignment Operators (=)

⇒ Assign values

int a = b;

chain ⇒ int a = b = c = 10; ✓

6. Ternary operator (? :)

↳ works on three operands.

a ~~? a~~ : a > b ? a : b

Operator Precedence.

Precedence means evaluation of operators based on precedence.

Highest

++ (postfix)					
++ (prefix)	--	~	!	+	-
*		%			
+					
>>	>>>	<<			
>	>=	<	<=		
**					
&					
^					
|					
&&					
||					
:					
->					
=					

Lowest

() → highest precedence BODMAS Rule.

int a = b + c * d ;

int a = (b + c) * d ;